

Foundation: ML Solutions for WSN Security

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BASE REFERENCE

Ghadi et al. (2024), IEEE Access

A comprehensive review of ML for Wireless Sensor Networks (WSN).

"The survey identifies four critical open issues: training location, privacy, lightweight algorithms, and trust/interpretability."

THE PROBLEM

Computational & Interpretability Gap

Standard high-accuracy models (e.g., Random Forest at 99.94%) are **computationally "heavy"** for resource-constrained sensor nodes.

These models lack **transparency**, creating a "black box" challenge for security trust.

CORE FINDINGS: CRITICAL RESEARCH DIMENSIONS

01

Training Location

02

Privacy

03

Lightweight Algorithms

04

Trust/Interpretability

The KB-Scale IDS Problem

Why WSN intrusion detection can't just import ML from the cloud.

THE HARDWARE

Sensor motes are tiny.

- TelosB-class motes: 10 KB RAM, 48 KB flash, 8 MHz MCU
- Battery-powered, no rechargeability in the field
- Every extra KB of model = fewer hours of operation
- Cluster heads in LEACH rotate — the detector has to fit on ANY node

THE MISMATCH

Strong IDS models are MB-scale.

- Current WSN-DS SOTA (Talukder 2025): Random Forest, ~85 MB on disk
- CNN/LSTM baselines: 5–50 MB
- None of them fit on a mote — they run at the base station
- Base-station-only detection = single point of failure

THE CORE CHALLENGE

We need a detector that matches cloud-grade accuracy on all 5 WSN-DS attack classes AND fits inside a single KB of flash.

Base Paper & the Unfilled Gap

Ghadi et al. 2024 – Comprehensive ML survey for WSN security.

BASE REFERENCE

Ghadi et al. 2024 (IEEE Access)

- Surveys ML approaches for Wireless Sensor Network security
- Identifies four open problems: training location, privacy, **LIGHTWEIGHT ALGORITHMS**, and trust/interpretability
- Explicit call for KB-scale models with explainable decisions
- Does NOT propose a compression pipeline — describes the gap

WHAT'S MISSING

No work closes lightweight + faithful XAI together.

- **Lightweight WSN-DS work exists but drops classes** (e.g., Alfarra 2025: 4 of 5)
- **XAI on WSN-DS exists** (Birahim 2025) but uses MB-scale ensembles
- **SOTA (Talukder 2025)**: 99.94% macro-F1 — but 85 MB, no XAI faithfulness check
- **No prior work audits** whether the COMPRESSED model's explanations still match the teacher's reasoning

OUR SLOT

Full 5-class WSN-DS + KB-scale footprint + SHAP faithfulness audit — **all three in one pipeline.**

How We Arrived at This Design

Every component has a defensible reason – including the ablation we expected might fail.

① WHY RF TEACHER

Tabular + imbalance + calibratable.

- WSN-DS is tabular and severely imbalanced (Normal dominates)
- Random Forests handle both natively, no feature engineering
- Can be calibrated with isotonic regression (Niculescu-Mizil 2005)
- Talukder 2025 confirms RF dominates the WSN-DS leaderboard

② WHY KNOWLEDGE DISTILLATION

The only way to hop architecture classes.

- Pruning and quantization stay inside the teacher's architecture
- KD (Hinton et al. 2015) lets us pick a radically smaller architecture
- Target is a ~1,200-parameter MLP — KD is the only lever
- Enables 85 MB → 1 KB shrinkage across architecture boundaries

③ WHY CL AS AN ABLATION

Pre-registered as a test, not a claim.

- Bengio 2009: easy→hard curricula can help imbalanced learning
- Wu et al. 2021 showed CL often does NOT help in standard settings
- We built CL as an honest ABLATION ARM — expected outcome uncertain
- Result is consistent with Wu et al.'s warning about supervised CL

④ WHY SHAP FAITHFULNESS AUDIT

Accuracy ≠ reasoning alignment.

- Compression is meaningless if the student is a black box
- Need a check that accuracy doesn't hide "learned via different features"
- Spearman rank correlation gives exactly that check
- This audit is what revealed our novel finding

CuKD-XAI Pipeline

Five stages, one audit loop.



SYSTEM PERFORMANCE & VALIDATION

18,335× on-disk compression (85 MB $\rightarrow$ 1.16 KB) + 5-seed Wilcoxon validation + SHAP audit on every config.

Experimental Setup

WSN-DS, 10 configs, 5 seeds, Wilcoxon signed-rank.

DATASET

WSN-DS (374,661 records)

- 5 attack classes: Normal, Blackhole, Grayhole, Flooding, Scheduling/TDMA
- LEACH protocol telemetry: Physical, MAC, and Network layer features
- Severe imbalance (Normal $\approx$ 90% of traffic)
- 80/20 stratified split, 5 seeds per config



Normal (340k) Attacks (34k total)

PROTOCOL

10 configs $\times$ 5 seeds

- **Baselines:** Random Forest, XGBoost, LightGBM, standalone MLP
- **Teachers:** B (standard MLP), C (CL MLP)
- **Students:** D, E (KD from RF), E2 (KD from MLP), F (KD from CL MLP)
- **Metrics:** macro-F1, per-class F1, ECE, FLOPs, on-disk size
- **Stats:** Wilcoxon signed-rank with Holm-Bonferroni correction
- **SHAP:** DeepExplainer on students, TreeExplainer on RF

Result 1 — Compression Works

Config E: KD from calibrated RF into a 1,189-parameter MLP.

ACCURACY

0.9219 ✓

macro-F1 (5 seeds, Wilcoxon-validated)

SIZE

1.16 KB

INT8 quantized student (on disk)

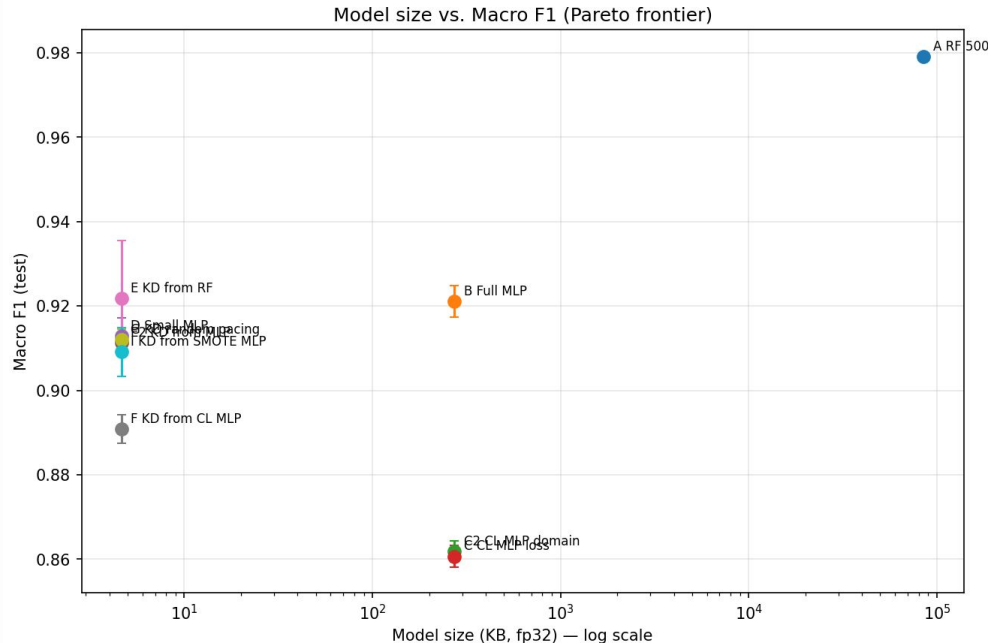
Parameters: 1,189 • FLOPs: ~2.4K per inference

COMPRESSION RATIO

18,335×

85 MB Random Forest → 1.16 KB INT8 MLP

Parameter ratio: 59× (69,893 → 1,189)



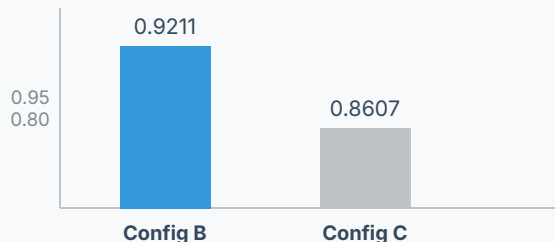
One calibrated-RF teacher + vanilla KD = cloud-grade accuracy at mote-scale footprint. Config E is our headline number.

Result 2 — CL Was Pre-Registered, and It Didn't Help

An honest negative – consistent with Wu et al. ICLR 2021.

TEACHERS

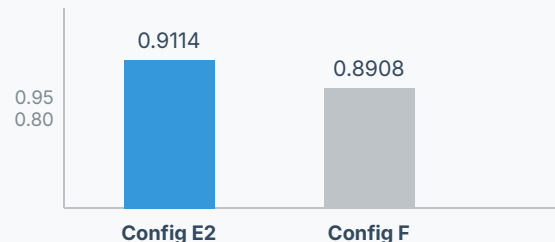
CL hurt the teacher by 6 points.



$\Delta = -0.0604$ macro-F1 (Wilcoxon $p < 0.01$)

STUDENTS

CL hurt the student by 2 points.



$\Delta = -0.0206$ macro-F1

WHY THIS IS GOOD NEWS

- We DID NOT hide this result — CL was pre-registered as an ablation.
- Wu et al. ICLR 2021 already showed CL often fails in supervised settings; our finding is consistent.
- The paper reports it as a rigorous negative result.
- Our **HEADLINE model (Config E)** stands independently at 0.9219 macro-F1.

Novel Finding — The Feature Alignment Gap

Compressed students learn the task, not the teacher's reasoning.

RF TEACHER SHAP

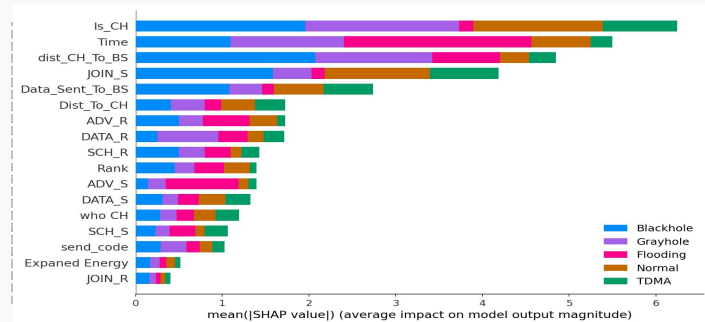
Top features the teacher relies on

INSERT teacher SHAP bar chart here — top-10 features ranked by mean |SHAP| value, TreeExplainer.

Spearman ρ
 ≈ 0
Rankings are effectively uncorrelated.

STUDENT SHAP

Top features the student relies on



Headline: KD transfers task accuracy but **NOT** feature-level reasoning. This is — to the best of our April 2026 literature search — an unreported phenomenon for tabular IDS compression. It is a novel contribution that holds regardless of whether the CL ablation had worked.

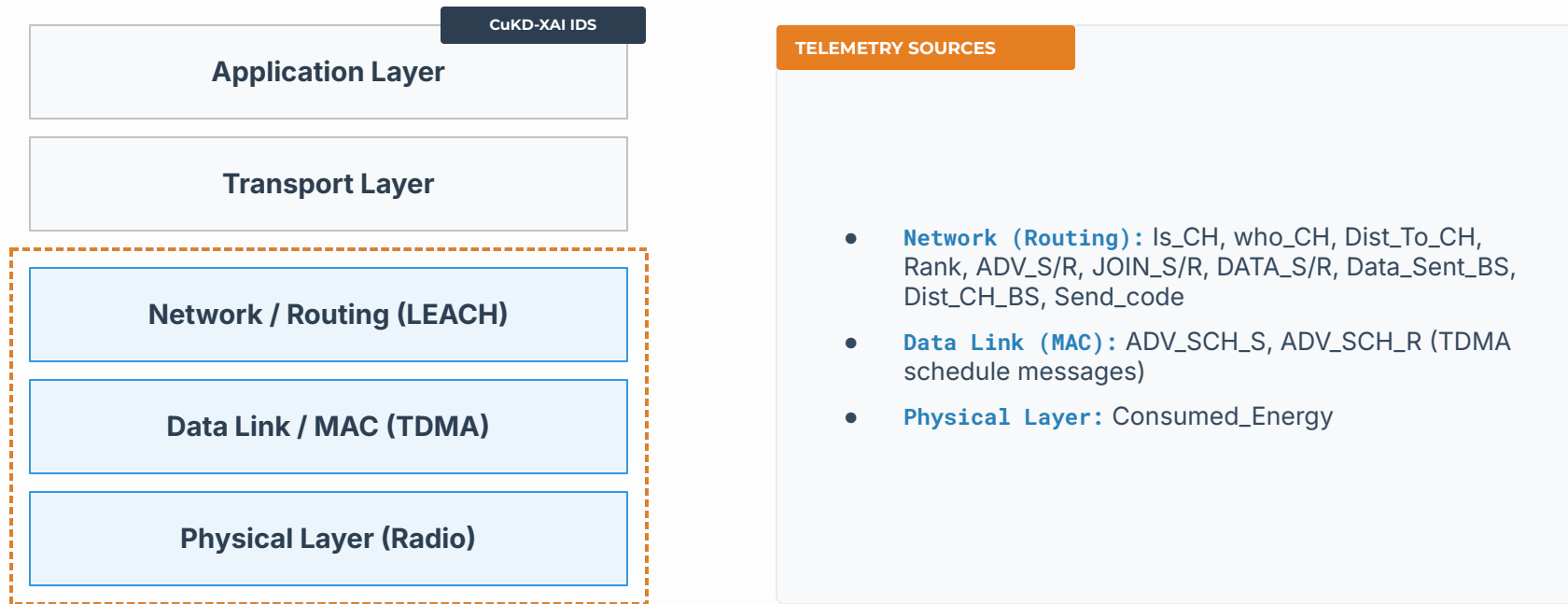
Where We Sit vs the 2025 Literature

Verified via CrossRef / OpenAlex, April 2026.

HEAD-TO-HEAD			
WSN-DS competitor landscape			
Paper	Full 5-class	KB-scale	SHAP faithfulness audit
Talukder 2025 (SOTA, 99.94%)	✓	✗ (85 MB RF)	✗
Alfarra 2025 (0.93, pruning)	✗ (4-class, drops TDMA)	✓	✗
Birahim 2025 (SHAP+LIME)	✓	✗ (MB-scale ensemble)	✗
Benaddi 2025 (KD+Kronecker)	✗ (TON_IoT, not WSN-DS)	✓	✗
CuKD-XAI (ours)	✓ (all 5 classes)	✓ (1.16 KB)	✓ (Spearman audit)

We are the only row with three green checks. Every other paper either drops classes, keeps MB-scale models, or skips the SHAP faithfulness audit.

Where CuKD-XAI Operates in the WSN Protocol Stack



"CuKD-XAI is a cross-layer IDS — it collects telemetry from Physical, MAC, and Network layers to detect attacks targeting the Network and MAC layers."

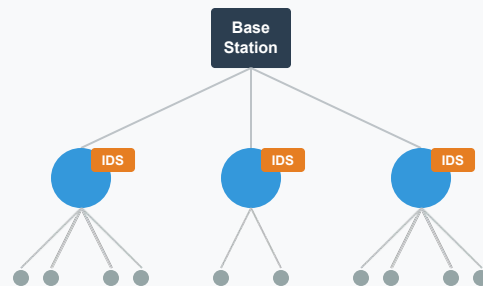
Attack Surface and Deployment Target

WSN Intrusion Detection: Threat Model & Topology Mapping

WSN-DS ATTACKS BY PROTOCOL LAYER

Attack	Layer	Mechanism
Blackhole	Network (Routing)	Drops all forwarded packets
Grayhole	Network (Routing)	Selectively drops packets
Flooding	Network (Routing)	Excess ADV/JOIN control messages during LEACH formation
Scheduling	Data Link / MAC	Manipulates TDMA time-slot assignment

DEPLOYMENT: IN-NODE DETECTION ON CLUSTER HEADS



- **Base Station** — centralized, single point of failure
- **Cluster Heads** — **our target** (1.16 KB INT8 student fits in 10 KB RAM of TelosB)
- **Regular Sensor Nodes** — feasible only through our KD compression

*KD compression (69,893 → 1,189 params, ~59×
enables application-layer IDS to run directly on resource-constrained sensor nodes instead of the base station.*